

BENNETT UNIVERSITY

School of Computer Science Engineering & Technology

MILESTONE 02

AI IN HEALTHCARE

Existing Systems, Feasibility & Proposed Methodology

MediGenius — AI-Powered Medical Information Assistant

CSET343 | AI in Healthcare Mini Project

SUBMITTED BY

Nidhi Kuntal | E23CSEU2357

Krish Rajpurohit | E23CSEU2366

1. Review of Existing Systems

MediGenius addresses a narrow but important gap: users want conversational explanations of health information, while clinical questions require source quality, uncertainty handling, and boundaries on what an automated system may safely claim. The review below evaluates relevant categories rather than treating all health platforms as direct competitors.

Existing system / approach	Established capability	Gap relevant to MediGenius
MedlinePlus (NLM) [2]	Reviewed health topics, medicines, tests and patient education.	High-quality reference material, but not a persistent, adaptive dialogue for a user's follow-up questions.
Ada symptom assessment [3]	Structured symptom input and probabilistic assessment of possible causes.	Different objective: symptom assessment; MediGenius is intentionally not a diagnostic or likelihood-ranking tool.
General-purpose LLM chat	Flexible natural-language explanations and conversational follow-up.	Answers can be fluent without traceable support; generic generation is unsafe for high-risk requests.
Document-based RAG [4]	Retrieves passages before generating an answer, improving source traceability.	Retrieval alone cannot verify clinical claims or ensure appropriate refusal and escalation.

1.1 Research Gap and Project Positioning

No single design choice guarantees trustworthy medical responses. The proposed contribution is the integration of evidence retrieval, conditional multi-agent orchestration, session-aware memory and explicit pre-/post-generation safety stages. Its value is architectural control and auditable behavior, not a claim of greater diagnostic accuracy than clinically evaluated systems.

MediGenius therefore targets health education, symptom-description preparation, and general medication explanations. It excludes diagnosis, prescriptions, dosage recommendations, interaction clearance, and autonomous emergency triage.

2. Project Feasibility

Feasibility is evaluated against the actual prototype architecture documented in the project repository [1], with unresolved concerns identified separately from implemented components.

Dimension	Evidence / feasibility judgement	Constraint or mitigation
Technical	Python/FastAPI backend, React frontend, LangGraph workflow, ChromaDB and SQLite are compatible with a local full-stack prototype.	Availability and configuration of Groq/Tavily services; implement graceful fallback.
Data and knowledge	A medical-reference PDF can be processed and indexed without collecting patient records.	Source currency and quality require documented provenance and periodic review.
Operational	Docker and automated tests support reproducible development and demonstrations.	API quotas, external latency and partial outages must be assessed repeatedly.
Safety and ethics	Rule-based refusal, claim verification and reviewer flags are implementable.	Rule coverage is incomplete by nature; clinician-reviewed evaluation remains necessary.
Privacy	Session history can be stored in SQLite with separated semantic-memory retrieval.	Minimize stored content, restrict cross-session access and define retention/deletion rules.

2.1 Feasibility Decision

Feasible as a supervised academic mini-project and non-clinical information assistant. Not established as safe for independent medical decision-making. Software test coverage, a working demo, or isolated response-time measurements cannot substitute for clinical validation [5].

3. Objectives of the Proposed Work

- O1 — Build a natural-language interface that explains general medical concepts using retrievable supporting material.
- O2 — Route requests by intent and risk, including emergency-language detection and explicit refusal of high-risk medication guidance.
- O3 — Combine PDF-based retrieval with relevant external knowledge only when a query benefits from additional evidence.
- O4 — Retain useful context within a user session without leaking conversation content across sessions.

- O5 — Check generated medical claims against available evidence and withhold or qualify insufficiently supported answers.
- O6 — Evaluate retrieval relevance, evidence support, refusal behavior, latency and software reliability on a defined test set.

4. Proposed Methodology

The methodology separates deterministic risk control from probabilistic language generation. The initial safety decision is made before expensive retrieval or LLM calls; a further verification stage examines the candidate answer before display.

Stage	Input → operation → output
1. Safety screening	User text → emergency/crisis patterns and refused-topic rules → static referral or continue.
2. Context preparation	Allowed query + session ID → recent SQLite history and session-scoped semantic recall → context bundle.
3. Task routing	Query → supervisor/planner → symptom structuring, direct explanation or retrieval path.
4. Evidence retrieval	Medical PDF chunks in ChromaDB; concurrent Wikipedia/Tavily lookup when applicable → candidate evidence.
5. Response synthesis	Evidence + context → Groq-hosted LLM through LiteLLM gateway → draft answer.
6. Verification and delivery	Check unsupported high-risk claims, remove ungrounded dosage figures → answer, limitation notice and audit flag.

Where evidence is absent or contradictory, the target behavior is to state uncertainty or refer the user to a qualified professional, rather than manufacture a definitive clinical conclusion.

5. Relevance of Algorithms and Techniques

Technique	Why it is used	Key limitation / safeguard
Sentence embeddings + cosine similarity	all-MiniLM-L6-v2 represents questions and text chunks semantically; ChromaDB retrieves the closest passages.	Semantic similarity is not clinical correctness; inspect retrieved evidence and handle low-quality hits.
Recursive text chunking + RAG	Segments long PDF content for retrieval, then conditions the response on retrieved evidence.	Chunk boundaries can omit context; preserve provenance and avoid unsupported synthesis.
LangGraph StateGraph + conditional routing	Makes distinct paths for safety, symptoms, retrieval, synthesis and verification explicit and testable.	Agent complexity adds latency and failure points; keep routing deterministic where possible.
Concurrent retrieval + timeout/fallback	Consults local knowledge, Wikipedia and Tavily without waiting sequentially for every source.	Live-web material may be variable; rank sources and handle failed branches.
Rule-based safety + LLM verification	Uses hard rules for refusals, followed by evidence-based review of generated claims.	Pattern matching misses paraphrases; LLM verification can also make errors.
SQLite + session-scoped vector memory	Supports multi-turn continuity while retrieving only context for the current session.	Requires access controls, data minimization and session-isolation testing.
RxNorm/RxNav name normalization	Recognizes medication names consistently before routing sensitive requests to a pharmacist.	Name recognition is not interaction checking or dosage validation.

6. Evaluation Design and Acceptance Criteria

A pre-defined, de-identified prompt suite will include ordinary health education, follow-up questions, irrelevant queries, emergency/crisis language, pediatric or pregnancy dosing, drug interactions and source-conflicting cases. The following are proposed evaluation measures, not reported experimental results.

Test area	Observable measure	What constitutes acceptable behavior
Evidence quality	Manual grounded-claim score; retrieved-passage relevance.	Clinical statements are linked to supporting passages; unsupported statements are flagged.
Safety behavior	Unsafe-compliance count, correct referral count, missed-risk cases.	No actionable dose/interaction clearance in disallowed cases; review false negatives.
Conversation integrity	Multi-turn coherence and cross-session leakage tests.	Follow-up context remains useful, with no cross-session content exposure.
Reliability	Repeated latency runs, error rate, fallback tests.	Failures return transparent limitations rather than invented answers.

Evaluation should report both successes and failures, with denominators and example cases. Repository claims about coverage or latency must be rerun independently before being used as final measured findings.

7. Milestone 2 Outcome

The review establishes that an educational, retrieval-grounded and safety-aware assistant is a realistic project scope. The methodology specifies the evidence path, control flow, memory boundary and verification checks; the algorithm selection is justified against explicit requirements. The next milestone should focus on reproducible implementation, controlled testing and documented failure analysis rather than clinical-performance claims.

References

- [1] <https://github.com/nidhi752/Al-in-Healthcare>.
- [2] U.S. National Library of Medicine, “MedlinePlus: Health Information,” <https://medlineplus.gov/>.
- [3] Ada Health, “How do I start a symptom assessment?” <https://ada.com/help/how-do-i-start-a-symptom-assessment/>.
- [4] P. Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” NeurIPS, 2020.
<https://proceedings.neurips.cc/paper/2020/hash/6b493230205f780e1bc26945df7481e5-Abstract.html>
- [5] World Health Organization, “Ethics and governance of artificial intelligence for health: WHO guidance,” 2021.
<https://www.who.int/publications/i/item/9789240029200>